



ROOT CAUSE UNLOCKED · CONDITION KIT

The Lyme Root Cause *Kit*

A Root Health Condition Guide

When a negative test means something, when it does not, what to do in the 72 hours after a tick bite, and why persistent symptoms are real without the testing being able to prove them.

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The Lyme Root Cause Kit

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BEFORE YOU START

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Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

Seek medical care now for facial droop, severe headache with neck stiffness, palpitations or fainting, or new severe joint swelling. Lyme can affect the nerves, heart, and joints, and those presentations need prompt treatment rather than a testing strategy.

This is the most consequential guide in this kit, because Lyme is a real bacterial infection where **delay changes outcomes** and where a large market of unproven treatment sits alongside the real one.

After treatment, when symptoms persist

This section was missing, and for a meaningful share of readers it is the section they need most.

The situation has a name and it is recognized. Post-treatment symptoms following Lyme disease sit within a broader group of infection-associated chronic conditions that also includes post-COVID conditions and myalgic encephalomyelitis. A 2026 review in the infectious disease literature argues for recognizing these conditions, notes that millions of Americans are affected, and calls for better surveillance, research, and inclusive trial design (PMID 41967005).

That matters because being told your symptoms are unexplained, after documented treatment for a documented infection, is its own injury. The condition is real, it is increasingly acknowledged, and the honest state of things is that treatment is not settled.

Pacing is the practical centerpiece. A scoping review of pacing in ME/CFS describes it as a prominent energy-management strategy while stating plainly that the literature is not yet sufficient to dictate treatment practice (PMID 37838675). Co-production work with adults living with ME/CFS puts energy management at the center of daily life, alongside sleep, diet, and psychological support (PMID 39961545).

Naming the population honestly: that research is in ME/CFS rather than in post-treatment Lyme. It is included because the clinical problem overlaps and because the safety principle is the same one used everywhere on this shelf: **if your symptoms reliably worsen a day or two after exertion, graded pushing is the wrong instruction.**

What that looks like in practice.

- Find your envelope on an average day, not a good one, and plan to the average.
- Break activity into shorter blocks with rest between, rather than one long push.
- Treat sleep as a treatment. Fixed window, seven to nine hours.
- Five to ten minutes daily of slow breathing with a longer exhale, for the nervous system load that chronic illness carries.
- Expect a non-linear course, and judge progress in months rather than days.

And the thing that has to be said plainly. Persistent symptoms after treatment are not evidence that you need indefinite antibiotics. Long-term antibiotic therapy for persistent symptoms has real harms, and that decision belongs with an infectious disease physician rather than with a protocol.

If you were bitten in the last few days, read this section first

There is a specific, evidence-backed action, and it has a time window.

A systematic review and meta-analysis of antibiotic prophylaxis after tick bite pooled 6 studies covering 3,766 people. Unfavorable events occurred in **0.4% of those treated versus 2.2% of controls**, a pooled risk ratio of **0.38**.¹

The subgroup analysis is the useful part. **Single-dose 200 mg doxycycline had the strongest result, at a risk ratio of 0.29.** A 10-day antibiotic course was not statistically significant, and topical azithromycin was not either. The authors concluded the evidence supports antibiotic prophylaxis and shows an advantage for the single dose.

So: after a qualifying tick bite, a single dose of doxycycline is the intervention with the best evidence, and it is time-limited. That is a same-week conversation with a clinician, not something to research for a month. Whether you qualify depends on tick species, attachment time, and local infection rates, which is exactly what the visit is for.

Do not wait for a test to decide. Antibodies take weeks to develop. Testing in the days after a bite tells you about past exposure, not about this bite.

The single most important testing principle

A rash that looks like erythema migrans is a clinical diagnosis. It does not require a positive blood test, and treatment should not wait for one.

This is where the most avoidable harm happens. Someone presents early, the serology is negative because antibodies have not developed yet, and treatment is deferred. The negative result was not evidence of no infection; it was evidence that not enough time had passed.

Why: serologic tests detect your immune response, not the organism. Early in infection that response has not yet formed. **A negative test early in the illness does not exclude Lyme disease.**

That principle is why the guideline-based approach treats a characteristic rash in an endemic area on clinical grounds.

What the standard testing actually is

Two-tier serology has been the standard: a first-tier immunoassay, then confirmation. **Modified two-tier testing**, using two immunoassays rather than an immunoblot, is now in use and has been compared directly against the standard algorithm.² Single-tier approaches are under development.³

What the tests do well: later-stage disease, where the antibody response is established.

What they do poorly: the first weeks, for the reason above.

What they cannot do at all: tell you whether an infection has been cured, or diagnose persistent symptoms after treatment. More on that below.

The authoritative reference is the 2021 clinical practice guideline developed jointly by the Infectious Diseases Society of America, the American Academy of Neurology, and the American College of Rheumatology.^{4,5} If you want one document behind your care, that is it.

Persistent symptoms after treatment, taken seriously and described accurately

Most people recover fully. The prognosis after appropriate antibiotic treatment of early or late Lyme disease is favorable.⁶

Some do not, and that is a recognized phenomenon. Post-treatment Lyme disease syndrome describes fatigue, musculoskeletal pain, and cognitive complaints persisting **six months or longer** after completing antibiotic therapy.⁶ A 2022 review reaches the same position: most patients fully recover with recommended therapy, while some report persisting nonspecific symptoms.⁷

Known risk factors include delayed diagnosis, greater symptom severity, and neurologic symptoms at the time of initial treatment.⁶ The first of those is the argument for not deferring treatment while waiting on a test.

And this is the part that matters for your labs: two-tier serologic testing is **neither sensitive nor specific for diagnosing PTLDS**, because antibody responses vary a great deal after treatment of early disease.⁶

Read that carefully in both directions. A positive antibody test months later does not prove ongoing infection, because antibodies persist after successful treatment. A negative one does not prove your symptoms are not real. **Serology cannot answer the question you most want answered**, and any test being sold as though it can is overreaching.

Optimal treatment of PTLDS awaits better understanding of the underlying mechanism.⁶ That is an unsatisfying sentence and it is the accurate one.

A caution about specialty testing and treatment

There is a documented market here. A study in Clinical Infectious Diseases set out specifically to identify and characterize the range of **unorthodox alternative therapies marketed to treat Lyme disease**, noting that patients with medically unexplained symptoms or alternative diagnoses who believe they have chronic Lyme are commonly targeted by providers of these therapies.⁸

I am not telling you your symptoms are imagined. I am telling you that a market exists which is built on the gap between real persistent symptoms and the absence of a proven treatment, and that being in that gap makes people vulnerable.

Practical filters for any test being offered:

- Is it performed by a lab accredited for this purpose?
- Has the assay been validated against a reference standard, or does its interpretive framework exist only in the marketing?
- If the result is positive, what specifically changes about my care?
- Does the person offering the test also sell the treatment?

And one thing to do regardless. Tell your physician everything you are taking. Complementary and alternative medicine use is common, reported by 36.7% of American adults in 2022, and **most patients do not disclose it to their physicians**, who may then underestimate the biological potency of what is being taken.⁹ With Lyme in particular, where people often run herbal antimicrobial protocols alongside prescription treatment, non-disclosure is a genuine safety issue rather than a courtesy problem.

Reading your results as a pattern

Characteristic rash, any test result. Clinical diagnosis. Treat.

Recent bite, no symptoms, negative test. Expected. The test is measuring the wrong thing this early. The prophylaxis conversation is the useful one.

Positive two-tier, compatible symptoms, exposure history. Treat per guideline.

Positive antibodies months after treated Lyme, ongoing symptoms. Antibodies persist after cure. This does not establish ongoing infection, and PTLDS is not diagnosed serologically.⁶

Negative standard testing, positive specialty-lab result, no exposure history. Apply the filters above before building a treatment plan on it.

Persistent symptoms, everything negative. Real symptoms, unproven cause. Worth pursuing the rule-outs that produce the same picture, including thyroid disease, anemia, sleep apnea, and vitamin deficiencies.

What to ask your clinician

- “I was bitten recently. Do I qualify for single-dose doxycycline prophylaxis?”
- “I have a rash. Does it need a positive test before treatment?”
- “My symptoms started weeks ago and the test was negative. Should it be repeated now that time has passed?”
- “I finished treatment and still feel unwell. What does the evidence say about what happens next?”
- “Here is everything I am taking, including supplements and herbs.” Bring the actual labels.

When to seek clinical review

Urgently: facial droop, severe headache with neck stiffness, palpitations, fainting, or new severe joint swelling.

This week: any recent tick bite, because prophylaxis is time-limited.

Promptly: an expanding rash after possible exposure.

Track what changes

Root Tracker: [rootcauseunlocked.com/tracker?](https://rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=lyme-lab)

[utm_source=guide&utm_medium=ebook&utm_campaign=lyme-lab](https://rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=lyme-lab)

Free, works anywhere. Where symptoms persist and testing cannot settle the question, a symptom record over months is genuinely one of the more useful things you can bring to an appointment.

Working with Root Health

Book a free Root Discovery Session: [rootcauseunlocked.com/discovery?](https://rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=lyme-lab)

[utm_source=guide&utm_medium=ebook&utm_campaign=lyme-lab](https://rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=lyme-lab)

Consultations are available to people located in North Carolina, where I am licensed. **If you have an active or suspected acute infection, that belongs with a physician promptly rather than with me.** I will say so.

The keys this kit follows

Unlock Your Lyme works through a set of root causes rather than a single organism, and this kit follows the same order: why the standard test fails, the co-infection picture, the right testing framework, the gut-Lyme connection, drainage first, diet and environment, and the autoimmune and neurological tail.

The reason that structure matters here more than in most conditions: Lyme is the condition where people most often chase the organism for years while the drivers that keep them unwell, sleep debt, an unrepaired gut, a nervous system stuck in threat, and an activity pattern that guarantees crashes, go unaddressed. Killing something is not the same as recovering.

References

1. Antibiotic prophylaxis for prevention against Lyme disease following tick bite: an updated systematic review and meta-analysis. *BMC Infect Dis*. 2021. PMID 34749665.
2. Performance of two modified two-tier algorithms for the serologic diagnosis of Lyme disease. *J Clin Microbiol*. 2024. PMID 38597655.
3. A novel single-tier serologic test to diagnose all stages of Lyme disease. *J Clin Microbiol*. 2025. PMID 40833084.
4. Clinical Practice Guidelines by the Infectious Diseases Society of America (IDSA), American Academy of Neurology (AAN), and American College of Rheumatology (ACR): 2020 Guidelines for the Prevention, Diagnosis and Treatment of Lyme Disease. *Clin Infect Dis*. 2021. PMID 33483734.
5. Same guideline, neurology publication. *Neurology*. 2021. PMID 33257476. *A published erratum exists for this version. It is a correction notice, not a retraction, and the guideline stands.*
6. Posttreatment Lyme disease syndrome. *Infect Dis Clin North Am*. 2016. PMID 25999226.
7. Persistent Symptoms After Treatment of Lyme Disease. *Infect Dis Clin North Am*. 2022. PMID 36116839.
8. Lantos PM, et al. Unorthodox alternative therapies marketed to treat Lyme disease. *Clin Infect Dis*. 2015. PMID 25852124.
9. Risks Associated With Undisclosed Complementary and Alternative Medicine in Clinical Care. *Mayo Clin Proc*. 2026. PMID 41770189.
10. Sanal-Hayes NEM, McLaughlin M, Hayes LD, et al. A scoping review of ‘Pacing’ for management of Myalgic Encephalomyelitis/Chronic Fatigue Syndrome (ME/CFS). *J Transl Med*. 2023;21(1):720. PMID 37838675
11. Thornton EJ, Hayes LD, Goodwin DS, et al. Managing Energy, and Shaping Care: Insights from Adults with Myalgic Encephalomyelitis/Chronic Fatigue Syndrome Through Co-Production Workshops. *Am J Med*. 2025;138(6):1001-1009. PMID 39961545
12. Iskander JK, Haridopolos S. Making Invisible Illnesses Visible: Recognizing and Responding to Infection-Associated Chronic Conditions. *Clin Infect Dis*. 2026;83(1):e58-e61. PMID 41967005

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Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

Nothing here replaces antibiotic treatment. Lyme is a bacterial infection and delaying treatment to try a dietary approach is the most consequential mistake available in this condition, because delayed diagnosis is a documented risk factor for persistent symptoms afterward. The lab guide in this kit covers that, along with the time-limited action after a tick bite.

Why this guide is not called “The Lyme Diet.” I looked for evidence that a particular diet affects Lyme outcomes and did not find it. What I did find is a real, specific, well-evidenced dietary job that Lyme patients face more than almost any other outpatient group: **recovering from the antibiotics.**

The thing that makes Lyme different

Lyme patients take antibiotics. Often for weeks, sometimes in repeated or extended courses. That is appropriate treatment and it also has a measurable cost that gets very little attention.

Antibiotics reshape the gut microbiome, and recovery takes months rather than days. In a study using shotgun metagenomics, 12 healthy men were followed over **six months** after a four-day antibiotic intervention. The initial changes included blooms of enterobacteria and other pathobionts including *Enterococcus faecalis* and *Fusobacterium nucleatum*, alongside **depletion of Bifidobacterium species and butyrate producers**.¹

Four days of antibiotics. Six months of follow-up. That is the scale of what a multi-week Lyme course is doing.

Two things follow.

The gut symptoms that appear during or after Lyme treatment are frequently the antibiotics, not the infection and not a new mystery illness. That reframing alone spares people a lot of worry and a lot of unnecessary testing.

And the useful dietary work is largely about protecting the microbiome during treatment and rebuilding it afterward. That is concrete, evidenced, and time-bound, which is more than most condition diets can say.

During treatment

Probiotics, which have the strongest number in this guide

Two independent meta-analyses agree closely. One pooled 42 studies with 11,305 participants and found co-administration of probiotics with antibiotics **reduced the risk of antibiotic-associated diarrhoea in adults by 37%**, a risk ratio of 0.63 with a confidence interval of 0.54 to 0.73.² Another pooled 36 studies with 9,312 participants and found a **38% reduction**, a pooled relative risk of 0.62 with a confidence interval of 0.51 to 0.74, with the protective effect holding across reasons for antibiotic treatment, duration, and dose.³

Two large analyses, different study sets, effectively the same answer. That is about as good as supplement evidence gets.

Practically: start when the antibiotics start rather than after problems appear, separate the probiotic from the antibiotic dose by a couple of hours, use named strains at stated counts, and continue for a period after the course finishes.

One caution worth stating: if you are immunocompromised, live probiotic organisms are a genuine clinical question. Ask first.

Food during the course

Eat, even when appetite drops. Under-eating during treatment slows everything downstream.

Keep fiber in. It feeds the butyrate producers that the antibiotics are depleting.¹ If bloating makes that hard, go with cooked vegetables and soluble sources rather than a raw salad.

Fermented foods if tolerated. Yogurt, kefir, sauerkraut, kimchi.

Take doxycycline with a full glass of water and stay upright afterward, because it can irritate the oesophagus, and be aware that calcium, iron, and magnesium reduce its absorption. Separate mineral supplements and large dairy servings from the dose. Doxycycline also causes photosensitivity, which is a sunburn risk rather than a diet issue but the most common surprise of the course.

Alcohol off during treatment. It burdens the same organ clearing the medication and does you no favours.

After treatment: rebuilding

The rebuild is where diet does its real work, and the timeline from the microbiome study is the honest expectation: this is months, not a fortnight.¹

Fiber, increased gradually and from varied sources. Variety matters more than volume. Vegetables, legumes, whole grains, fruit, nuts, seeds. Increase slowly, because a fast jump after antibiotics produces bloating that gets blamed on the food.

Fermented foods regularly, not occasionally.

Protein sufficient for recovery, at every meal.

Polyphenol-rich foods. Berries, olive oil, herbs, tea, cocoa.

Rebuild the plate rather than the pill count. The microbiome study shows depletion across whole functional groups, and no capsule replaces a varied diet.

What I would not do: start a restrictive elimination diet during or immediately after treatment. Your microbiome is already depleted and restriction narrows it further at exactly the wrong moment.

For persistent symptoms

If fatigue, pain, and cognitive complaints continue six months or more after adequate treatment, the lab guide covers what that is and what testing can and cannot say about it.

From a food standpoint, the useful targets are the ones that mimic it. Iron, B12, and vitamin D deficiency each produce fatigue and cognitive fog and each is testable and correctable. Persistent symptoms after Lyme get attributed to the infection when a deficiency is sitting untreated underneath, and that is a genuinely common miss.

Blood sugar stability matters for the day-to-day pattern. Protein at every meal, no liquid sugar, and a short walk after the largest meal do more for afternoon crashes than any protocol.

Eating for sleep, because sleep loss lowers pain thresholds and this population is often in a pain-and-poor-sleep loop. Caffeine earlier, alcohol not used as a sleep aid, and the last large meal away from bedtime.

A Mediterranean-style pattern is a reasonable default. It is not a Lyme treatment, and I am not going to present it as one.

Running a fair experiment

Do the rebuild first, before you experiment with restriction. Give it eight to twelve weeks.

Then one change at a time, with a written baseline, because gradual change is invisible from the inside.

Give restrictions an endpoint. If eight weeks produced nothing, stop. Permanent restriction without benefit is the most common self-inflicted harm in this space, and stacking eliminations on a depleted microbiome makes it worse.

Track it: rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=lyme-diet

Realistic expectations

No diet treats *Borrelia*. Antibiotics do, and most people fully recover with recommended therapy.

Probiotics alongside antibiotics reduce antibiotic-associated diarrhoea by roughly 37 to 38% across two large meta-analyses.^{2,3} That is the strongest claim in this guide and it is about tolerating treatment, not curing infection.

Microbiome recovery after antibiotics runs on a months-long clock.¹ Expecting to feel rebuilt in two weeks sets you up to abandon something reasonable.

And where symptoms persist, the highest-value dietary work is correcting the deficiencies that produce the same picture, which is testing followed by food rather than a protocol.

When to seek clinical review

Urgently: facial droop, severe headache with neck stiffness, palpitations, fainting, new severe joint swelling. Also severe or bloody diarrhoea during or after antibiotics, which can indicate *C. difficile* and is a medical problem rather than a dietary one.

Promptly: inability to keep food or fluids down during treatment.

Before a restrictive diet: if you are underweight, have a history of disordered eating, are pregnant, or are managing diabetes.

Working with Root Health

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Consultations are available to people located in North Carolina, where I am licensed. **If you have an acute or suspected active infection, that belongs with a physician promptly**, and I will say so rather than take the appointment.

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The nervous system and the stress axis

This belongs in a Lyme kit for a specific reason: sustained illness, contested diagnosis, and months of uncertainty are a genuine physiological load, not a soft one.

What I will claim and what I will not. There is no trial showing that stress reduction clears an infection, and I am not going to imply one. What the literature on infection-associated chronic conditions and on fatigue-dominant illness does consistently include is psychological support as a component of comprehensive care alongside sleep, diet, and energy management (PMID 39961545, PMID 41967005).

Why it matters practically. A nervous system held in threat degrades sleep, and poor sleep worsens pain, cognition, and fatigue. That loop is enterable from either end, and the end you can reach without a prescription is the daily one.

The work:

- Five to ten minutes daily of slow breathing, longer exhale than inhale. Consistency beats technique.
- A fixed sleep window, protected like a medication.
- Pacing, covered in the lab guide, which is the single most important behavioral instruction in this condition.
- Eat sitting down and unhurried, because digestion is parasympathetic and the gut work in this kit depends on it.

And the thing worth saying out loud. Being disbelieved is part of this illness for many people, and it is exhausting in its own right. That is a reason to bring someone competent alongside you, not a reason to conclude the symptoms are imaginary.

References

1. Palleja A, et al. Recovery of gut microbiota of healthy adults following antibiotic exposure. *Nat Microbiol*. 2018;3(11):1255-1265. PMID 30349083.
2. Goodman C, et al. Probiotics for the prevention of antibiotic-associated diarrhoea: a systematic review and meta-analysis. *BMJ Open*. 2021. PMID 34385227.
3. Probiotics for the Prevention of Antibiotic-associated Diarrhea in Adults: A Meta-Analysis of Randomized Placebo-Controlled Trials. *J Clin Gastroenterol*. 2021. PMID 33234881.
4. Thornton EJ, Hayes LD, Goodwin DS, et al. Managing Energy, and Shaping Care: Insights from Adults with Myalgic Encephalomyelitis/Chronic Fatigue Syndrome Through Co-Production Workshops. *Am J Med*. 2025;138(6):1001-1009. PMID 39961545
5. Iskander JK, Haridopolos S. Making Invisible Illnesses Visible: Recognizing and Responding to Infection-Associated Chronic Conditions. *Clin Infect Dis*. 2026;83(1):e58-e61. PMID 41967005

Doxycycline administration and interaction guidance reflects the product prescribing information.

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This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

Seek medical care now for facial droop, severe headache with neck stiffness, palpitations, fainting, or new severe joint swelling.

Read the lab guide first. It covers the single time-limited action that matters most, prophylaxis after a tick bite, and why a negative early test does not exclude infection.

Why nutrient evidence looks thinner than it is

A large outcome trial costs tens to hundreds of millions of dollars, and that money only exists when a patent can pay it back. You cannot patent magnesium, an herb, or a food. So the evidence hierarchy this guide grades against is not neutral: it systematically over-documents patentable drugs and under-documents nearly everything else. That is measurable, not rhetorical. A Cochrane methodology review found that manufacturer sponsorship of drug and device studies produces more favourable results and conclusions than independent funding, through a bias that standard quality checks do not catch (PMID 28207928).

Two honest consequences follow, and they pull in opposite directions.

First, “no evidence” has three different meanings, and this guide tries to always tell you which one applies. *Tested and failed:* a real trial ran and the answer was no. *Never properly tested:* nobody could afford the trial, so absence of evidence is not evidence of absence. *Tested small and won:* real but modest trials, usually measuring markers rather than outcomes, because markers are what a small budget can afford.

Second, the asymmetry earns nutrients humility, not automatic credit. When isolated nutrients have received large publicly funded trials, the results have been mixed: some genuine wins, some nulls, and some harms. Small nutrient trials also carry their own publication bias toward positive results, just as drug trials tilt toward their sponsor. So this guide applies one rule in both directions: name the funding, name the evidence tier, and let you see both.

The one instruction in this guide that is not optional

Tell your physician everything you are taking.

Complementary and alternative medicine use was reported by **36.7% of American adults in 2022, and most patients do not disclose it to their physicians**, who may consequently underestimate the biological potency of what is being taken.¹

In Lyme this is not a courtesy issue, it is a safety issue. People frequently run herbal antimicrobial protocols alongside prescription antibiotics. Herbs are pharmacologically active, some affect the same liver enzymes that clear medications, and some affect bleeding. Your prescriber cannot account for what they do not know about.

Bring the actual labels, not product names, because blends contain things the name does not mention.

Where the honest boundary sits

Lyme disease is a bacterial infection. Its treatment is antibiotic, guided by a joint guideline from the Infectious Diseases Society of America, the American Academy of Neurology, and the American College of Rheumatology.² **Most patients fully recover with recommended therapy.**³

No supplement treats the infection, and delaying antibiotic treatment to try one is the single most consequential mistake available in this condition, because delayed diagnosis is a documented risk factor for persistent symptoms afterward.⁴

So what is a supplement guide for here? Two legitimate things: supporting general recovery, and helping you evaluate the very large market you are about to encounter. The second is worth more.

The market, described accurately

A study published in Clinical Infectious Diseases set out specifically to identify and characterize the range of **unorthodox alternative therapies marketed to treat Lyme disease**. Its framing is worth quoting in substance: patients with medically unexplained symptoms or alternative diagnoses who believe they chronically have Lyme are **commonly targeted by providers of alternative therapies**.⁵

Here is why that market exists, and I want to be fair to everyone in it. **Post-treatment Lyme disease syndrome is real**. Fatigue, musculoskeletal pain, and cognitive complaints persisting six months or longer after adequate antibiotic therapy is a described phenomenon with recognized risk factors.⁴ And there is no proven treatment for it; optimal management awaits better understanding of the mechanism.⁴

A real condition with no proven treatment is exactly the space where an industry grows. That does not make everyone in it dishonest, and it does mean the burden of proof is on the seller.

Filters worth applying to anything offered to you:

- Does the person recommending it also sell it?
- Is the claimed mechanism testable, or does worsening also count as evidence it is working?
- Is there an endpoint, or is this indefinite?
- What would have to happen for them to tell you to stop?

That third one catches most of it. A protocol with no defined endpoint is a subscription.

On prolonged antibiotic therapy specifically. Extended antibiotic courses for post-treatment symptoms have been the subject of controlled trials, and the literature on them is limited and has been reviewed.⁶ This is a decision for a physician working from the guideline,² not something to pursue through a supplement protocol or an out-of-network prescription without that framing. Long-term intravenous antibiotics carry real risks including line infections.

What is reasonable

The unglamorous foundation, which is most of what is defensible

Sleep, protein sufficient for recovery, and correcting documented deficiencies in iron, B12, and vitamin D. The fatigue and cognitive symptoms attributed to persistent Lyme overlap almost exactly with those deficiencies, with thyroid disease, and with sleep apnea. Those are testable and treatable, and they are in the rule-out list in the lab guide.

That is not an exciting recommendation. It is the one with evidence behind it.

Pacing, if post-exertional crashes are part of your picture

Where activity reliably produces a delayed crash, pace by what you can repeat rather than by what you can manage on a good day. This costs nothing and prevents the boom-and-bust pattern that turns a slow recovery into a flat one.

Probiotics during and after antibiotics

Reasonable, low risk, and worth using for the antibiotic-associated gut effects rather than as anything anti-Borrelia. Named strains, and separate from the antibiotic dose by a couple of hours.

Ask your practitioner first

No products and no links here.

Any herbal antimicrobial protocol, if you are on prescription treatment. This is the interaction question above, and it is the most likely place for real harm. *Ask:* “Here are the labels of everything I am taking. Does any of it interact with my treatment or affect my liver enzymes?”

Anything at all, if you take an anticoagulant. Several popular botanicals affect bleeding.

Prolonged or intravenous antibiotics. A physician decision, made against the guideline, with the risks stated.^{2,6}

Any therapy involving a device, an infusion, or a procedure. Those are the highest-cost and highest-risk end of the market described above,⁵ and they deserve the filters.

What I deliberately left out, and why

The herbal antimicrobial protocols. They are the centrepiece of the alternative Lyme market. I searched for clinical trial evidence in patients and did not find it. Laboratory activity against *Borrelia* is not the same as a human outcome.

Anything sold as eradicating persistent infection. Whether persistent infection causes PTLDS is an open scientific question, not a settled premise you can buy a solution to.

Protocols with die-off or Herxheimer framing. A model in which feeling worse confirms the treatment is working cannot be tested against reality, and unfalsifiable is a warning sign rather than a feature.

Things I use in practice that are not in here. A guide sold to someone I have never examined has no history, no medication list, and no way to change course. Where the evidence does not support recommending something at a distance, it stays out. In this condition more than any other in the kit, I would rather send you to a physician than sell you a plan.

Where to find quality versions of these

Everything above was written without reference to any product line, and this is the guide with the least to buy.

Third-party testing first. NSF Certified for Sport and USP Verified.

us.fullscript.com/welcome/drseay

Disclosure: Root Health earns a portion of sales through that link. Nothing here was included, excluded, or ranked based on what it earns, and the largest-selling category in the Lyme space is the one this guide declines to recommend.

When to seek clinical review

Urgently: facial droop, severe headache with neck stiffness, palpitations, fainting, new severe joint swelling.

This week: any recent tick bite. Prophylaxis is time-limited and covered in the lab guide.

Before starting anything: if you are on prescription treatment, an anticoagulant, or have liver disease.

Working with Root Health

Book a free Root Discovery Session: [rootcauseunlocked.com/discovery?
utm_source=guide&utm_medium=ebook&utm_campaign=lyme-supplement](https://rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=lyme-supplement)

Consultations are available to people located in North Carolina, where I am licensed. **If you have an acute or suspected active infection, that belongs with a physician promptly.** I will tell you so rather than take the appointment.

If symptoms persist after treatment

Read the full section in **Understanding Your Lyme Labs**, because it is the part of this condition where the wrong instruction does the most harm. The short version:

Post-treatment symptoms sit within the recognized group of infection-associated chronic conditions (PMID 41967005). The practical centerpiece is pacing rather than pushing. A scoping review of pacing in myalgic encephalomyelitis and chronic fatigue syndrome describes it as a prominent energy-management strategy while stating that the literature is not yet sufficient to dictate practice (PMID 37838675), and co-production work with adults living with ME/CFS puts energy management at the center of daily life alongside sleep, diet, and psychological support (PMID 39961545).

That research is in ME/CFS, not post-treatment Lyme, and I would rather name the population than imply a match. The reason it applies is the shared safety principle used across this shelf: if symptoms reliably worsen a day or two after exertion, graded pushing is the wrong instruction.

No supplement in this guide treats that. Sleep, pacing, and a daily down-regulating practice do more for it than anything you can buy, and persistent symptoms are not a reason for indefinite antibiotics, which is an infectious disease physician's decision.

The keys this kit follows

Unlock Your Lyme works through a set of root causes rather than a single organism, and this kit follows the same order: why the standard test fails, the co-infection picture, the right testing framework, the gut-Lyme connection, drainage first, diet and environment, and the autoimmune and neurological tail.

The reason that structure matters here more than in most conditions: Lyme is the condition where people most often chase the organism for years while the drivers that keep them unwell, sleep debt, an unrepaired gut, a nervous system stuck in threat, and an activity pattern that guarantees crashes, go unaddressed. Killing something is not the same as recovering.

The nervous system and the stress axis

This belongs in a Lyme kit for a specific reason: sustained illness, contested diagnosis, and months of uncertainty are a genuine physiological load, not a soft one.

What I will claim and what I will not. There is no trial showing that stress reduction clears an infection, and I am not going to imply one. What the literature on infection-associated chronic conditions and on fatigue-dominant illness does consistently include is psychological support as a component of comprehensive care alongside sleep, diet, and energy management (PMID 39961545, PMID 41967005).

Why it matters practically. A nervous system held in threat degrades sleep, and poor sleep worsens pain, cognition, and fatigue. That loop is enterable from either end, and the end you can reach without a prescription is the daily one.

The work:

- Five to ten minutes daily of slow breathing, longer exhale than inhale. Consistency beats technique.
- A fixed sleep window, protected like a medication.
- Pacing, covered in the lab guide, which is the single most important behavioral instruction in this condition.
- Eat sitting down and unhurried, because digestion is parasympathetic and the gut work in this kit depends on it.

And the thing worth saying out loud. Being disbelieved is part of this illness for many people, and it is exhausting in its own right. That is a reason to bring someone competent alongside you, not a reason to conclude the symptoms are imaginary.

References

1. Risks Associated With Undisclosed Complementary and Alternative Medicine in Clinical Care. *Mayo Clin Proc.* 2026. PMID 41770189.
2. Clinical Practice Guidelines by the Infectious Diseases Society of America (IDSA), American Academy of Neurology (AAN), and American College of Rheumatology (ACR): 2020 Guidelines for the Prevention, Diagnosis and Treatment of Lyme Disease. *Clin Infect Dis.* 2021. PMID 33483734.
3. Persistent Symptoms After Treatment of Lyme Disease. *Infect Dis Clin North Am.* 2022. PMID 36116839.
4. Posttreatment Lyme disease syndrome. *Infect Dis Clin North Am.* 2016. PMID 25999226.
5. Lantos PM, et al. Unorthodox alternative therapies marketed to treat Lyme disease. *Clin Infect Dis.* 2015. PMID 25852124.
6. Controlled trials of antibiotic treatment in patients with post-treatment chronic Lyme disease. *Vector Borne Zoonotic Dis.* 2004. PMID 12804167.
7. Sanal-Hayes NEM, McLaughlin M, Hayes LD, et al. A scoping review of 'Pacing' for management of Myalgic Encephalomyelitis/Chronic Fatigue Syndrome (ME/CFS). *J Transl Med.* 2023;21(1):720. PMID 37838675
8. Thornton EJ, Hayes LD, Goodwin DS, et al. Managing Energy, and Shaping Care: Insights from Adults with Myalgic Encephalomyelitis/Chronic Fatigue Syndrome Through Co-Production Workshops. *Am J Med.* 2025;138(6):1001-1009. PMID 39961545
9. Iskander JK, Haridopolos S. Making Invisible Illnesses Visible: Recognizing and Responding to Infection-Associated Chronic Conditions. *Clin Infect Dis.* 2026;83(1):e58-e61. PMID 41967005
10. Lundh A, Lexchin J, Mintzes B, et al. Industry sponsorship and research outcome. *Cochrane Database Syst Rev.* 2017;2(2):MR000033. PMID 28207928

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